巴适声工厂隐私版使用手册 / Bashi Voice Factory Privacy Edition User Guide

Bashi Voice Factory Privacy Edition (巴适声工厂 · 隐私版)

Version: 0.1.6

A local desktop web app for text-to-speech and speech-to-text. Synthesis and transcription use local models without uploading text or audio to cloud speech services. Prepare the required assets online before offline use. Access defaults to this computer; optional LAN access transfers data between devices over plain HTTP.

Author: Alex Li (ncorecpu@gmail.com) **License:** MIT License **Source code:** https://github.com/gtree965/bashi-voice-factory-privacy · https://gitee.com/gtree965/bashi-voice-factory-privacy (mirror for mainland China) **Contributing:** https://github.com/gtree965/bashi-voice-factory-privacy/blob/main/CONTRIBUTING.md **Download:** GitHub Releases · Gitee (recommended in mainland China)

What's New in v0.1.6

- One-time LAN PIN with lockouts, Host and cross-site request checks; plain HTTP still requires a trusted network. See security boundaries and fixes.
- Dependency-source testing and fallback, with HTTPS certificate verification restored. The launcher shows a 20-second local-only default prompt and a more useful LAN address.
- Bilingual privacy and offline-verification instructions ship with the package. The manual's extracted text and links are checked before packaging.

What's New in v0.1.5

- **Complete third-party license materials now ship inside the package:** `THIRD_PARTY.md`, `THIRD_PARTY_GAPS.md`, and the entire `licenses/` directory — 34

files in total — including the upstream Qwen3-TTS-GGUF conversion runtime's MIT license text.
- **UI footer license wording corrected:** it now reads "project code: MIT; models and third-party components follow their respective license terms" and no longer makes a blanket license promise for the whole distribution.
- **Export filenames corrected.**
- **Quick Start launch instructions corrected.**
- **Build-time license pack gate added:** the license texts inside every release package are verified against the repository.

What's New in v0.1.4

- **Faster first synthesis:** full-path TTS warmup now starts asynchronously when the UI loads, using a throwaway synthesis rather than model loading alone. The UI shows a neutral warming state; an early request waits for warmup instead of failing.
- **Fixed the PDF user guide:** the bundled manual no longer carries broken internal links that pointed at directories on the build machine. All links in the manual are now public HTTPS addresses.
- **Build-time guard:** packaging now refuses to ship a user guide containing any non-HTTPS link, so this class of defect cannot recur.
- **Gitee mirror:** source and downloads are now also served from Gitee for users in mainland China.

Highlight Features

Fully Local Text-to-Speech

- **Local Qwen3-TTS-12Hz-1.7B-CustomVoice** running entirely on your machine — no cloud calls during synthesis.
- **9 curated voice presets** with native-language sample previews built in.
- **10 languages** supported: Chinese (Mandarin, Cantonese, Sichuanese, Beijing, Northeast), English (US/UK), Japanese, Korean, German, French, Russian, Portuguese, Spanish, Italian.
- **Instruction control** for style, accent, emotion via natural-language prompts.
- **Streaming controls for long text and Shadowing Mode**: stop synthesis while keeping completed chunks, auto-play the first ready chunk by default, and continue playback as later chunks arrive; GGUF long text also preserves a complete seekable/downloadable MP3 after the streamed preview.

Automatic Backend Selection

- **GGUF + Vulkan** for AMD / Intel / iGPU users (default, fast, RAM-friendly).
- **GGUF + CUDA** for NVIDIA users via a one-click in-app upgrade (optional ~595 MB add-on download; no manual weight setup). New in v0.1.1.

- **Reliable NVIDIA detection and startup probes**: v0.1.2 detects the physical NVIDIA GPU through virtual-display layers and isolates native probe crashes so the launcher can report an actionable error instead of disappearing.
- **CPU fallback** for entry-level hardware (works on Intel N100-class boxes).
- One-click **speed test** calibrates ETA estimates to your specific machine.

Local Offline Speech-to-Text (STT)

- **SenseVoice Small** (default, 242 MB) — multilingual zh/en/ja/ko/yue, non-autoregressive (fast on CPU).
- **Parakeet TDT 0.6B** (optional, 661 MB) — English specialist, NVIDIA's ~1.7% WER champion.
- **VAD-based segmentation** via Silero — accurate timestamps, no chunk-boundary stutter.
- **Live transcript streaming** via SSE; exports to TXT / SRT / VTT.

Smart First-Launch Download with Resume

- **HTTP Range / resume** built in — a transient WiFi drop won't restart your 2.2 GB GGUF download from zero.
- **Recovery on failure**: package-source fallback for dependencies; resume and backoff for model downloads.
- **Download-source selection**: in Chinese language/time-zone environments, pip sources are tested and ranked, with fallback on failure; synthesis assets use ModelScope and transcription models use hf-mirror.com with upstream fallback.

LAN Sharing for Phone / Tablet Use

- At start-up the launcher asks whether other devices on your network may connect (20 seconds, local-only by default).
- If you allow it, the console shows the address to open, for example `http://192.168.1.52:5050`, and a one-time six-digit PIN that other devices must enter first.
- The connection is plain HTTP: only enable it on a network you trust. See SECURITY.md for what the PIN does and does not protect.

Quick Start

Easiest path

System prerequisites: 64-bit Windows, Windows long-path support, and the Microsoft Visual C++ x64 runtime. A fresh Windows or isolated environment may need that runtime installed separately; the portable ZIP does not include it. If the launcher reports disabled long-path support, follow the instructions it prints before retrying. Prepare these system requirements before disconnecting.

1. Unzip the downloaded file to any folder (Desktop or wherever you like).
2. **Double-click** `Start_启动.bat` at the top level of the extracted folder.
3. First launch will:
   - Install Python dependencies (~700 MB, 8-15 min depending on network/CPU)
   - Automatically download the GGUF model when missing (~2.2 GB, 2-15 min depending on network; Ctrl+C cancels and the next launch resumes)
   - When startup finishes, the console window shows the address; open `http://127.0.0.1:5050` yourself in the browser of your choice (the app does not open a browser automatically)
4. With dependencies and the models you need prepared, core tasks can run offline. Download the transcription model and VAD in Model Manager before disconnecting; repair and optional downloads still need a connection.

Advanced

Users who prefer to bypass the top-level launcher can run `bashi-privacy-app\run_portable.bat` directly. Both paths are identical.

Network Behavior

Synthesis and transcription use local models and do not upload text or audio to cloud speech services. Default access is local-only; in LAN mode, data travels between devices over plain HTTP.

- The launcher may install or repair dependencies, bootstrap pip and download missing synthesis assets. With a Chinese language or time zone, it tests Aliyun, Tsinghua, Tencent Cloud, Huawei Cloud and PyPI, then installs in ranked order with fallback; elsewhere it uses PyPI. Built-in HTTPS sources retain certificate checks. A custom `BASHI_PIP_INDEX_URL` skips testing and an HTTP override has no HTTPS protection.
- Transcription-model downloads use `hf-mirror.com`, then `huggingface.co`; downloading a model also prepares missing Silero VAD from GitHub. The optional speaker-label models are hidden by default and downloaded from GitHub only when requested.
- The optional NVIDIA CUDA add-on is downloaded from ModelScope when you request the upgrade. The update button opens `https://gitee.com/gtree965/` in a browser tab. There is no telemetry or background update polling.
- Prepare synthesis assets, the transcription model and VAD before disconnecting. Optional downloads and repair require a connection; a later launch is not unconditionally offline.

The privacy policy lists the download sources, fallback and override boundaries. The offline verification method includes helper scripts and a blank record template; it does not claim any package has passed. For isolated use, prepare the needed assets online, then follow that method. TCP polling and proxy observations have limits and are not packet captures.

Hardware Expectations

The table below shows **actually measured** numbers from author hardware. The built-in speed test (click 测速 / Speed Test on the right-hand panel) calibrates the ETA panel to your specific machine on first use, so you don't have to extrapolate from this table.

Hardware	Auto-selected backend	"你好。" probe (25 chars)	1,000-char synthesis ETA	Status
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	~3 s	a few minutes	Tested 2026-05
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 s	~5-10 min	Tested
Intel N305 laptop + UHD iGPU	GGUF + Vulkan / DirectML	53 s	25-46 min	Tested 2026-05-25
Intel N100 mini-PC + UHD iGPU	GGUF + Vulkan / DirectML	126 s	58 min - 1h49 min	Tested 2026-05-25
NVIDIA RTX 5070 (12 GB, cloud PC, Blackwell)	GGUF + Vulkan/CUDA + DirectML	~1 s	41 s - 1 m 17 s	Tested 2026-06-12 (Vulkan) & 2026-07-24 (海马云 HMv Cloud PC): full cold-start synthesis confirmed; optional CUDA add-on verified working (LLM 28 layers on CUDA0, decoder DirectML fp16). CUDA-vs-Vulkan A/B still welcome.

Hardware	Auto-selected backend	“你好。” probe (25 chars)	1,000-char synthesis ETA	Status
Other NVIDIA RTX / GTX (desktop)	GGUF + Vulkan + DirectML · optional CUDA add-on	community reports welcome	community reports welcome	awaiting desktop tester reports via GitHub Issues
Apple Silicon / Intel Arc	—	not yet measured	not yet measured	not validated yet

> **Entry-level CPUs (Intel N100 / N305 class) are only practical for short text — under ~200 characters per generation.** A 5,000-character essay would take 2-9 hours on these boxes. For long-form audio (lectures, audiobooks), please use discrete-GPU hardware (AMD RX 500/600/9000 series, NVIDIA RTX class).

> ℹ **NVIDIA users (desktop RTX / GTX)**: the default path is GGUF + Vulkan (NVIDIA cards support Vulkan via the proprietary driver). For native CUDA acceleration, click the one-click in-app upgrade banner (v0.1.1+) — it downloads a ~595 MB CUDA runtime add-on, no manual weight setup required. **Requires NVIDIA driver ≥ 545.x for the CUDA 12.4 runtime.** Vulkan path measured on RTX 5070 (cloud PC): **1 s for the 25-char probe**, so the default Vulkan experience is already excellent on a desktop NVIDIA card. CUDA add-on A/B numbers + reports from other RTX/GTX cards welcome via GitHub Issues. Cloud datacenter NVIDIA cards (A10/A100/T4) running in TCC mode have additional setup steps — see Troubleshooting below.

Hardware Coverage Matrix

Beyond the specific machines benchmarked above, here is how the auto-selected acceleration path maps across mainstream Windows hardware:

Hardware class	Acceleration path used	Status
NVIDIA RTX 30 / 40 / 50 (desktop)	GGUF + Vulkan default · optional CUDA in-app upgrade	RTX 5070 (Blackwell) verified on 海马云 cloud PC — 2026-06-12 Vulkan probe ~1 s; 2026-07-24 full cold-start synthesis + optional CUDA add-on confirmed working (LLM 28 layers on CUDA0, decoder DirectML fp16, driver 610.47). CUDA-vs-Vulkan A/B numbers + other-card reports still welcome via Issues. CUDA add-on requires driver ≥ 545.x.
NVIDIA GTX 10 / 16 (desktop)	GGUF + Vulkan default · optional CUDA in-app upgrade	Same flow, same driver requirement; reports welcome via Issues.
NVIDIA datacenter (A10 / A100 / T4)	Manual setup required	TCC mode + old cloud driver = manual workaround. See Troubleshooting.
AMD RX 500 / 600 / 7000 / 9000 (discrete)	GGUF + Vulkan + DirectML	Tested (RX 590, RX 9060 XT)
Intel Arc A-series (A380 / A580 / A750 / A770)	GGUF + Vulkan + DirectML	Should work — not yet measured
Intel iGPU (UHD / Iris Xe / Arc iGPU)	GGUF + Vulkan + DirectML	Tested (Intel N305 UHD)
AMD APU iGPU (Vega 7/8, RDNA 2/3)	GGUF + Vulkan + DirectML	Should work — not yet measured
CPU only (no usable GPU / no driver)	GGUF + CPU (ggml-cpu auto-SIMD)	Works; auto-selected only when no GPU detected
NPU (Snapdragon X / Lunar Lake / Ryzen AI 300)	—	Not yet utilized
ARM64 Windows (Snapdragon X Copilot+ PC)	—	Not supported (runs via slow x64 emulation)

Known Limitations (roadmap)

- **NVIDIA CUDA is opt-in, not bundled.** By default NVIDIA cards use the Vulkan path; native CUDA acceleration requires the one-click in-app upgrade (~595 MB add-on, v0.1.1+). This keeps the main download small for the AMD / Intel / CPU majority. Requires NVIDIA driver ≥ 545.x.
- **Weak iGPU may be slower than pure CPU.** On Intel N100-class hardware, DirectML / Vulkan transfer + scheduling overhead can outweigh GPU benefit. A/B test by launching from a cmd window after `set GGUF_LLM_USE_GPU=0 && set GGUF_ONNX_PROVIDER=CPU` — feedback on real speed numbers welcome.
- **NPU acceleration is not yet used** (Snapdragon X Elite, Intel Lunar Lake AI Boost, AMD Ryzen AI 300). Under investigation for v0.2+.
- **ARM64 Windows is not natively supported** (Surface Pro 11, Galaxy Book4 Edge, etc.). x64 emulation works but is slow. Native ARM64 build is a v0.3+ candidate.

Troubleshooting

Symptom	Likely cause / fix
`Start_启动.bat` shows "Array index expression is missing"	Old zip — re-download the latest from GitHub Releases or Gitee; the BOM bug was fixed in v0.1.0 final
pip install stops mid-way after WiFi blip	The launcher retries or switches package sources. If installation still fails, fix the connection and restart; completed installs can be reused from cache.
pip install fails with a "long path support" message	Run the registry one-liner from the launcher's advisory in an Administrator PowerShell, then re-launch
pip install fails on every mirror, or a mirror refuses your IP (`403` / `denied by IP ACL`)	The launcher ranks five built-in indexes in Chinese environments and switches on failure; elsewhere it uses PyPI. To force one index, set `BASHI_PIP_INDEX_URL` before launching, e.g. `set BASHI_PIP_INDEX_URL=https://pypi.tuna.tsinghua.edu.cn/simple/`
GGUF download interrupted	Same: retries with HTTP Range/resume; re-run launcher to continue from `.part` file

Symptom	Likely cause / fix
App exits with “No usable backend was found”	Check `app.log` (structured, timestamped application logs) and `launch_log.txt` (launcher steps and raw stderr) — usually GGUF runtime DLL missing, GPU driver outdated, or RAM <8 GB. App now prints a bilingual advisory with specific causes
STT download says “镜像失败”	Should not happen in v0.1.0 final — old behavior from removed ModelScope path
Cloud / datacenter NVIDIA GPU (A10 / A100 / T4 on Chinese cloud Windows images): `access violation reading 0x0000000000000000` or `GGUF probe failed`	Datacenter NVIDIA GPUs typically run in **TCC mode** (Vulkan/DirectML disabled) with older drivers (~538.x) incompatible with the CUDA 12.4 add-on. **Desktop RTX/GTX cards are NOT affected** (they run WDDM mode by default with modern drivers). In v0.1.2, native startup-probe crashes are isolated and reported instead of terminating the launcher, but TCC setup still requires the manual workaround: (1) rename `vulkan_backend_spike\Qwen3-TTS-GGUF\qwen3_tts_gguf\inference\bin\ggml-vulkan.dll` to `.disabled`; (2) edit `bashi-privacy-app\run_portable.ps1` and add `$env:USE_GGUF_BACKEND = "1"` + `$env:GGUF_ONNX_PROVIDER = "CPU"` after the `Remove-Item Env:USE_GGUF_BACKEND` lines (~line 270); (3) pre-install the CUDA add-on via CLI `python download_cuda_runtime.py`. CUDA 12.4 requires NVIDIA driver ≥ 545.x. Automatic TCC handling remains planned for a later patch.

Full log paths: `bashi-privacy-app\app.log` and `bashi-privacy-app\launch_log.txt`

What’s in the Zip

```
bashi-voice-factory-privacy-v0.1.6/
├── Start_启动.bat                                  ← double-click here
├── Start_CPU_only_仅CPU启动.bat                    ← force CPU mode
(entry-level iGPU A/B)
```

```
├── README.md                                          ← this file
├── README_CN.md                                       ← Chinese version
├── LICENSE                                            ← MIT license
├── VERSION                                            ← release version
├── PRIVACY.md / SECURITY.md                           ← network and
security boundaries
├── docs/OFFLINE_VERIFICATION.md                       ← method and blank
record template
├── 巴适声工厂隐私版使用手册_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf
│                                                       ← help PDF
(bilingual)
├── bashi-privacy-app/                                 ← app code +
embedded Python
│   ├── run_portable.bat                               ← same launcher,
direct path
│   └── ...
└── vulkan_backend_spike/                              ← GGUF runtime
(populated on first launch)
    └── Qwen3-TTS-GGUF/
```

License

MIT License © 2026 Alex Li

Bundled third-party components retain their original licenses:

- **Qwen3-TTS-12Hz-1.7B-CustomVoice**: Tongyi Lab (Alibaba), Apache 2.0
- **GGUF runtime**: based on HaujetZhao/Qwen3-TTS-GGUF
- **SenseVoice Small / Silero VAD / Parakeet TDT**: see respective model cards

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Issues, feedback, and feature requests welcome via GitHub Issues.

巴适声工厂 · 隐私版 (Bashi Voice Factory Privacy Edition)

版本： 0.1.6

在自己电脑上运行的语音工具：合成与转写使用本地模型，不向云端语音服务上传文本或音频。断网使用前须联网备齐所需资产。默认仅本机访问；主动开启局域网后，数据会经明文 HTTP 在设备之间传输。

作者： Alex Li (ncorecpu@gmail.com) **许可：** MIT License **源码：** https://github.com/gtree965/bashi-voice-factory-privacy · https://gitee.com/gtree965/bashi-voice-factory-privacy（国内镜像） **贡献指南：** https://github.com/gtree965/bashi-voice-factory-privacy/blob/main/CONTRIBUTING.md **下载：** GitHub Releases · Gitee（国内推荐）

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v0.1.6 更新内容

- 局域网一次性 PIN、失败次数限制、Host 与跨站请求检查；HTTP 仍未加密，仅用于可信网络。见安全边界与修复。
- 依赖源测速排序和回退，恢复 HTTPS 证书校验；启动询问等待 20 秒，默认仅本机，改进局域网地址提示。
- 随包提供双语隐私说明和断网核查方法；出包前核对手册链接及提取后的可读文本。

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v0.1.5 更新内容

- **发行包内首次附带完整第三方许可材料：** `THIRD_PARTY.md`、`THIRD_PARTY_GAPS.md` 与 `licenses/` 整目录，共 34 份（含上游 Qwen3-TTS-GGUF 转换运行时的 MIT 原文）。
- **界面页脚许可表述修正：** 改为「项目代码 MIT；模型与第三方组件遵循各自许可条款」，不再对整个发行包作统一授权承诺。
- **导出文件名修正。**
- **快速开始的启动说明修正。**
- **构建期新增许可材料落包门禁：** 确保每次发行包内的许可原文与仓库一致。

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v0.1.4 更新内容

- **首次合成更快：** 界面加载时即异步预热完整 TTS 链路，用一次丢弃式合成代替单纯加载模型。界面显示中性的预热状态；预热期间的早到请求会等待，而不是直接失败。

- **修复 PDF 使用手册：** 随包分发的手册不再带有指向构建机器本地目录的失效内链，手册内所有链接均为公开 HTTPS 地址。
- **构建期门禁：** 打包时若手册中存在任何非 HTTPS 链接即拒绝出包，杜绝此类缺陷复发。
- **Gitee 镜像：** 源码与下载新增 Gitee 入口，方便中国大陆用户访问。

核心亮点

完全本地文字转语音

- **Qwen3-TTS-12Hz-1.7B-CustomVoice 本地推理** — 合成过程零云端调用。
- **9 个精选音色**，内置母语试听样例。
- **10 种语言**：中文（普通话、粤语、四川话、北京话、东北话）、英文（美/英）、日语、韩语、德语、法语、俄语、葡萄牙语、西班牙语、意大利语。
- **自然语言指令控制** — 风格、口音、情绪都可通过 instruct 提示调整。
- **长文与跟读流式控制** — 可停止合成并保留已完成片段；默认自动播放首个就绪片段，后续片段生成后自动续播；GGUF 长文试听结束后仍保留可拖动、可下载的完整 MP3。

后端自动选择

- **GGUF + Vulkan**：AMD / Intel / 集成显卡用户默认路线（速度快、内存友好）。
- **GGUF + CUDA**：NVIDIA 用户一键应用内升级（可选 ~595 MB 附加包下载，无需手动配置权重）。v0.1.1 新增。
- **更可靠的 NVIDIA 检测与启动探测**：v0.1.2 可穿透虚拟显示层识别真实 NVIDIA 显卡，并隔离原生探测崩溃，让启动器显示可处理的错误而不是直接消失。
- **CPU 回退**：入门级硬件（如 Intel N100 系列）也能跑。
- 一键 **测速** 校准 ETA 估算，让等待时间贴合你这台机器。

本地离线音视频转文字 (STT)

- **SenseVoice Small**（默认，242 MB）— 中英日韩粤多语种，非自回归架构，CPU 上速度快。
- **Parakeet TDT 0.6B**（可选，661 MB）— 英文专用，NVIDIA 出品，WER 约 1.7%。
- **Silero VAD 精准分段** — 时间戳准确、零重叠、零卡顿。
- **实时滚屏字幕** SSE 流式输出；支持导出 TXT、SRT、VTT。

智能首次下载，断点续传

- **HTTP Range / resume 内置** — 一次 WiFi 闪断不会让 2.2 GB 的 GGUF 下载从零开始。
- **失败重试与恢复** — 依赖安装失败时重试或换源；模型下载保留断点和指数退避。

- **下载源选择**：中国语言或时区环境下先测速排序，失败时换源；合成资产来自 ModelScope，转写模型优先 hf-mirror.com、失败后回退上游。

局域网共享，手机平板可访问

- 启动时询问是否允许局域网内其他设备连接（等待 20 秒，默认仅本机）。
- 允许后，控制台会显示访问地址（例如 `http://192.168.1.52:5050`）和一次性 6 位 PIN，其他设备须先输入 PIN。
- 链路是明文 HTTP，请只在信任的网络上开启。PIN 能挡住什么、挡不住什么，见 SECURITY.md。

快速上手

最简流程

系统前提： 64 位 Windows、Windows 长路径支持，以及 Microsoft Visual C++ x64 运行库。全新 Windows 或隔离环境可能需要单独安装此运行库，便携 ZIP 不包含它。若启动器提示长路径未启用，请先按控制台指引完成设置，再重试。断网前须备齐这些系统条件。

1. 把下载的 zip 解压到任意文件夹（桌面或随便哪里都行）。
2. 在解压后的顶层目录里 **双击** `Start_启动.bat`。
3. 首次启动会：
   - 安装 Python 依赖（约 700 MB，根据网速 / CPU 不同需要 8-15 分钟）
   - 缺少 GGUF 模型时自动下载（约 2.2 GB，2-15 分钟；可按 Ctrl+C 取消，下次启动会续传）
   - 启动完成后，命令行窗口会显示访问地址；请在你想用的浏览器里手动打开 `http://127.0.0.1:5050`（应用不会自动打开浏览器）
4. 依赖和所需模型备齐后，核心任务可断网使用。断网前须在模型管理器下载转写模型及 VAD；修复和可选下载仍需联网。

进阶

如果想跳过顶层启动器，可以直接运行 `bashi-privacy-app\run_portable.bat`，效果完全一样。

网络行为说明

合成与转写使用本地模型，不向云端语音服务上传文本或音频。默认仅本机访问；局域网模式下，数据会经明文 HTTP 在设备之间传输。

- 首次准备或依赖检查失败时，启动器可能安装、修复依赖和引导 pip；合成资产未齐时会下载。中国语言或时区环境先对阿里云、清华、腾讯云、华为云、PyPI 测速，再按结果安装和回退；其他环境用 PyPI。内置 HTTPS 源校验证书；自定义 `BASHI_PIP_INDEX_URL` 不测速，HTTP 覆盖源没有 HTTPS 保护。

- 用户下载转写模型时先尝试 `hf-mirror.com`，失败后用 `huggingface.co`；同时从 GitHub 准备缺失的 Silero VAD。说话人标注界面默认隐藏，相关模型只在用户请求时从 GitHub 下载。
- 可选 NVIDIA CUDA 包由用户点击升级后从 ModelScope 下载。“查看最新版本”只在浏览器打开 `https://gitee.com/gtree965/`；没有遥测或后台更新轮询。
- 断网前须准备好合成资产、转写模型及 VAD。可选下载和修复仍需联网，不能无条件承诺每次后续启动都离线。

隐私说明 列出下载来源、回退与覆盖设置的边界。断网验证方法 提供辅助脚本和空白记录模板，不宣称某个发行包已经通过。隔离环境使用前，先联网备齐所需资产，再按方法核查；TCP 轮询和代理观察均有盲区，也不是抓包。

硬件要求

下表是**作者实测**的数据。首次运行后点右侧面板的“测速”按钮，程序会根据你这台机器实际表现重新校准 ETA 估算 — 不用照搬此表。

硬件	自动选择的后端	“你好。”探测（25 字）	1,000 字合成预计	状态
AMD RX 9060 XT (16 GB)	GGUF + Vulkan	约 3 秒	几分钟	2026-05 实测
AMD RX 590 (8 GB)	GGUF + Vulkan	3-5 秒	约 5-10 分钟	实测
Intel N305 笔记本 + UHD 集显	GGUF + Vulkan / DirectML	53 秒	25-46 分钟	2026-05-25 实测
Intel N100 迷你主机 + UHD 集显	GGUF + Vulkan / DirectML	126 秒	58 分钟 - 1 小时 49 分	2026-05-25 实测

硬件	自动选择的后端	“你好。”探测（25 字）	1,000 字合成预计	状态
NVIDIA RTX 5070（12 GB，云电脑，Blackwell）	GGUF + Vulkan/CUDA + DirectML	约 1 秒	41 秒 - 1 分 17 秒	2026-06-12（Vulkan）& 2026-07-24（海马云 HMv Cloud PC）实测：全冷启动合成确认；可选 CUDA 附加包已实证可用（LLM 28 层跑 CUDA0、解码 DirectML fp16）。CUDA-vs-Vulkan A/B 仍欢迎反馈。
其它 NVIDIA RTX / GTX（桌面）	GGUF + Vulkan + DirectML · 可选 CUDA 附加包	待社区报数	待社区报数	等待桌面 tester 通过 GitHub Issues 报数
Apple Silicon / Intel Arc	—	暂未实测	暂未实测	尚未验证

> **入门级 CPU（Intel N100 / N305 一类）仅适合短句试用 — 单次生成建议不超过 200 字。** 这类机器跑 5,000 字长文需要 2-9 小时。如果想做长文音频（讲座、有声书），请用独立显卡硬件（AMD RX 500/600/9000 系、NVIDIA RTX 类）。

> ℹ **NVIDIA 用户（桌面 RTX / GTX）**：默认路线是 GGUF + Vulkan（NVIDIA 驱动支持 Vulkan）。想要原生 CUDA 加速，点击应用内一键升级横幅（v0.1.1+）— 会下载 ~595 MB CUDA 运行时附加包，无需手动配置权重。**需要 NVIDIA 驱动 ≥ 545.x 才能使用 CUDA 12.4 运行时。** Vulkan 路径在 RTX 5070（云电脑）实测：**25 字探测 1 秒**，桌面 NVIDIA 用户的默认 Vulkan 体验已经非常顺畅。CUDA 附加包 A/B 数字 + 其它 RTX/GTX 卡的报数欢迎通过 GitHub Issues 反馈。云端数据中心 NVIDIA 显卡（A10/A100/T4）在 TCC 模式下需要额外设置 — 见下方常见问题排查。

硬件兼容性矩阵

除了上表里具体测过的几台机器，下表说明主流 Windows 硬件在自动后端选择下走的加速路径：

硬件类别	实际加速路径	状态
NVIDIA RTX 30 / 40 / 50（桌面）	默认 GGUF + Vulkan · 可选 CUDA 应用内升级	RTX 5070（Blackwell）海马云云电脑已验证——2026-06-12 Vulkan 探测约 1 秒；2026-07-24 全冷启动合成 + 可选 CUDA 附加包实证可用（LLM 28 层跑 CUDA0、解码 DirectML fp16、驱动 610.47）。CUDA-vs-Vulkan A/B 数字 + 其它卡报数仍欢迎通过 Issues 反馈。CUDA add-on 需驱动 ≥ 545.x。
NVIDIA GTX 10 / 16（桌面）	默认 GGUF + Vulkan · 可选 CUDA 应用内升级	同一流程，同样驱动要求；报数欢迎通过 Issues。
NVIDIA 数据中心卡（A10 / A100 / T4）	需要手动配置	TCC 模式 + 云端老驱动 = 需要手动 workaround。见常见问题排查。
AMD RX 500 / 600 / 7000 / 9000（独立显卡）	GGUF + Vulkan + DirectML	实测（RX 590、RX 9060 XT）
Intel Arc A 系列（A380 / A580 / A750 / A770）	GGUF + Vulkan + DirectML	理论可用、未实测
Intel 集显（UHD / Iris Xe / Arc iGPU）	GGUF + Vulkan + DirectML	实测（Intel N305 UHD）
AMD APU 集显（Vega 7/8、RDNA 2/3）	GGUF + Vulkan + DirectML	理论可用、未实测
纯 CPU（无可用 GPU / 无驱动）	GGUF + CPU（ggml-cpu 自动 SIMD）	可用；仅在无 GPU 时自动选择
NPU（Snapdragon X / Lunar Lake / Ryzen AI 300）	—	暂未利用
ARM64 Windows（Snapdragon X Copilot+ PC）	—	暂不支持（走 x64 仿真较慢）

已知限制（路线图）

- **NVIDIA CUDA 为可选项，不内置。** NVIDIA 显卡默认走 Vulkan 路线；原生 CUDA 加速需点击应用内一键升级（~595 MB 附加包，v0.1.1+）。这样主下

载包对 AMD / Intel / CPU 大多数用户保持精简。需 NVIDIA 驱动 ≥ 545.x。

- **弱集显可能比纯 CPU 还慢。** Intel N100 类硬件上，DirectML / Vulkan 的数据传输与调度开销可能盖过 GPU 收益。可在 cmd 窗口里 `set GGUF_LLM_USE_GPU=0 && set GGUF_ONNX_PROVIDER=CPU` 后再启动来 A/B 测试，欢迎反馈实测数字。
- **NPU 加速尚未利用**（Snapdragon X Elite、Intel Lunar Lake AI Boost、AMD Ryzen AI 300）。v0.2+ 调研中。
- **ARM64 Windows 暂未原生支持**（Surface Pro 11、Galaxy Book4 Edge 等）。x64 仿真能跑但速度受限。原生 ARM64 构建是 v0.3+ 候选。

常见问题排查

现象	可能原因 / 处理
双击 `Start_启动.bat` 报 “Array index expression is missing”	旧版 zip。请从 GitHub Releases 或 Gitee 重新下载最新版；该 BOM 问题在 v0.1.0 正式版已修复
pip 安装途中 WiFi 闪断后中止	启动器会重试或切换依赖源。仍失败时修复连接后重启，已安装部分可从缓存复用。
pip 安装失败并提示 “long path support”	按启动器提示，在管理员 PowerShell 中运行一次注册表命令，然后重新启动
所有镜像都装不上，或镜像拒绝本机 IP（`403` / `denied by IP ACL`）	中国环境下先对五个内置源测速排序，失败时换源；其他环境用 PyPI。若要指定某一个源，启动前设置 `BASHI_PIP_INDEX_URL`，例如 `set BASHI_PIP_INDEX_URL=https://pypi.tuna.tsinghua.edu.cn/simple/`
GGUF 下载中断	同样自动重试，且支持 HTTP Range 续传，重新运行启动器从 `.part` 文件续传
启动报 “No usable backend was found”	查 `app.log`（带时间戳的结构化应用日志）和 `launch_log.txt`（启动器步骤与原生 stderr）。常见原因：GGUF 运行 DLL 缺失、显卡驱动过旧、可用内存 <8 GB。程序会打印中英双语提示告知具体原因
STT 下载闪过 “镜像失败”	v0.1.0 正式版不会出现 — 是旧版残留的 ModelScope 路径，已删除

现象	可能原因 / 处理
云端 / 数据中心 NVIDIA 显卡（A10 / A100 / T4，国内云 Windows 镜像）：`access violation reading 0x0000000000000000` 或 `GGUF probe failed`	数据中心 NVIDIA 显卡通常运行在 **TCC 模式**（Vulkan/DirectML 被屏蔽），且驱动版本较旧（~538.x）跟 CUDA 12.4 add-on 不兼容。**桌面 RTX/GTX 显卡不受影响**（默认 WDDM 模式 + 现代驱动）。v0.1.2 会隔离并报告原生启动探测崩溃，不再让启动器直接退出，但 TCC 配置仍需手动 workaround：(1) 把 `vulkan_backend_spike\Qwen3-TTS-GGUF\qwen3_tts_gguf\inference\bin\ggml-vulkan.dll` 重命名为 `.disabled`；(2) 编辑 `bashi-privacy-app\run_portable.ps1`，在 `Remove-Item Env:USE_GGUF_BACKEND` 行（约第 270 行）之后加上 `$env:USE_GGUF_BACKEND = "1"` + `$env:GGUF_ONNX_PROVIDER = "CPU"`；(3) 命令行预装 CUDA add-on：`python download_cuda_runtime.py`。CUDA 12.4 需要 NVIDIA 驱动 ≥ 545.x。TCC 自动处理仍计划在后续补丁实现。

完整日志路径：`bashi-privacy-app\app.log` 和 `bashi-privacy-app\launch_log.txt`

---

zip 包目录结构

```
bashi-voice-factory-privacy-v0.1.6/
├── Start_启动.bat                                          ← 双击这里
├── Start_CPU_only_仅CPU启动.bat                            ← 强制 CPU 模式（入门集显 A/B 对比用）
├── README.md                                               ← 英文文档
├── README_CN.md                                            ← 本文件
├── LICENSE                                                 ← MIT 许可证
├── VERSION                                                 ← 发布版本号
├── PRIVACY.md / SECURITY.md                                ← 网络与安全边界
├── docs/OFFLINE_VERIFICATION.md                            ← 方法与空白记录模板
├── 巴适声工厂隐私版使用手册_Bashi_Voice_Factory_Privacy_Edition_User_Guide.pdf
│                                                           ← 中英双语帮助 PDF
├── bashi-privacy-app/                                      ← 程序代码 + 嵌入式 Python
│   ├── run_portable.bat                                    ← 等效启动器（直接路径）
│   └── ...
└── vulkan_backend_spike/                                   ← GGUF 运行时（首次启
```

动自动填充）
    └── Qwen3-TTS-GGUF/

许可

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第三方组件保留各自原始许可：

- **Qwen3-TTS-12Hz-1.7B-CustomVoice**：通义实验室（阿里巴巴）Apache 2.0
- **GGUF runtime**：基于 HaujetZhao/Qwen3-TTS-GGUF
- **SenseVoice Small / Silero VAD / Parakeet TDT**：详见各自模型卡

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欢迎通过 GitHub Issues 提交 issue、反馈或功能建议。